Literature review — outline for editing

Planning document, not prose. For each work: the **one point** taken from it, the **milestone** it moves the argument to, and **where it bites** in the dissertation. Edit here; full text follows on approval.

The graph

Nine chains. Arrows are "this work creates the problem the next one answers".

```
A  Carney --> ECB --> Acharya et al.
  (why stress test)      (why the current ones fail: not auditable)
                                    |
                                    +--------------+
                                    v              v
B  Battiston --> Bolton&Kacperczyk   |        I  Kenyon et al. --> BKMN
  (network is necessary)            |           (auditable chain answers Acharya)
       +--> Desnos                  |                     |
            (scenario is random)    |                     |
                                    v                     |
C  Barrage&Nordhaus --> Pindyck / Dietz&Stern --> SwissRe   |
  (damage fn)   (it has no empirical basis)   (5x range)   |
       +--> ND-GAIN (ranks, does not measure)              |
       +--> NGFS (3 IAMs + NiGEM; NO price->sector map) ---+
                                    |                      |
                                    v                      |
D  Hafner --> Böhringer (CGE) / Mercure (macroecon)         |
  (4 families)  Dafermos (SFC) / Lamperti (ABM)            |
                (what each omission costs)                 |
                                    |                      |
                                    v                      |
E  Leontief'36 --> Hawkins&Simon --> Leontief'70 --> Miller&Blair
  (the system)     (when it exists)  (emissions in)  (dual, blocks, aggregation)
                                    |                      |
                                    v                      |
F  Isard --> Chenery/Moses --> Leontief&Strout --> Stone     |
  (exact, unusable) (weaken)   (estimate flows)  (fit them) |
       +--> Tukker&Dietzenbacher --> Timmer / Moran&Wood --> OECD
            (measure them instead)   (databases agree ~10%)  |
       +--> Davis&Caldeira (trade weight != embodied emissions)
                                    |                      |
                                    v                      |
G  Weitzman'74 --> Green --> Kay&Jolley --> Roncalli&Semet  |
  (instrument)    (modest)  (IO+NGFS, 1 country) (incidence open)
       +--> RBB / Weber&Wasner (pass-through varies, can exceed 1)
       +--> Böhringer again (leakage 5-20%, BCA is a carbon tariff)
                                    |                      |
                                    v                      v
H  Moessner vs Konradt&Weder / Bauer&Känzig --> Taylor --> Hull&White --> Sarno&Taylor
  (inflation: DISPUTED)                        (rule)     (curve)      (parities)
                                                                        |
                                                                        v
                                                           THE GAP: nothing joins
                                                           BKMN's chain to the MRIO
                                                           apparatus
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A. Why stress test at all, and why the existing ones fall short

Milestone reached: climate is a prudential problem; the exercises that answer it are credible but unauditable.

Work	Point taken	Where it bites
Carney (2015)	*Tragedy of the horizon* — costs fall beyond every relevant planning horizon; remedy is forward-looking scenario analysis	§1 opening motivation
ECB (2022)	The largest implementation; establishes the standard shape: external scenario -> exposure map -> propagate -> revalue. Also the object Acharya et al. criticise, so it must be cited for that critique to be checkable	§1; the shape our chain also follows
Acharya et al. (2023)	Four criticisms. Transition risk treated as exogenous when it is a policy choice; no climate–economy feedback; no compound scenarios; **and not reproducible by the tested institution**	§1 (why auditability matters), §5 conclusion

Edit note: the fourth criticism is the one that justifies the whole BKMN design. Consider whether to lead with it rather than list all four.

B. Which network does a shock travel through?

Milestone reached: indirect exposure must be computed through an explicit network — but *which* network is open.

Work	Point taken	Where it bites
Battiston et al. (2017)	Indirect exposure via counterparties ~ direct exposure; affected loan share ~ bank capital. Networks must be computed, not inferred	§1, §2 — our argument for an IO network is theirs with a different network
Bolton & Kacperczyk (2023)	Carbon premium exists across 14,000 firms / 77 countries; larger where policy stricter -> transition risk **already partly priced**	§3.5 equity beta caveat; §5 limitation
Desnos et al. (2023)	Turns a deterministic stress test into stochastic climate VaR: a scenario is a *draw*, not a state	§3.1 — the direct antecedent for the Dirichlet mixture

Edit note: Desnos is from Roncalli's group — worth saying, since it links strand B to strand G.

C. Where scenarios come from, and the damage-function fight

Milestone reached: the damage function is the least-evidenced object in the chain, and NGFS hands users a scenario with no sectoral cost map.

Work	Point taken	Where it bites
Barrage & Nordhaus (2024)	DICE itself — Ramsey growth, carbon cycle, quadratic damage — and the DICE-2023 recalibration supplying the coefficient used	§3.4 the Ω form; §4.1
Pindyck (2013)	No theoretical or empirical basis for the curvature at high T; outputs are illustrations, not estimates. Sole carrier of the functional-form criticism	§5 limitation; justifies carrying two calibrations
Dietz & Stern (2015)	Damage on capital/growth rather than level -> SCC far higher	§5 — our damage is a *level* effect only
Swiss Re Institute (2024)	Coefficient ~5× DICE at any temperature	§3.4 second calibration; §4 sensitivity
ND-GAIN (2024)	Ranks vulnerability; does **not** measure loss — must be paired with a damage function	§3.4 σ_r; §4.1
NGFS (2024)	3 IAMs (GCAM, MESSAGEix, REMIND) + NiGEM; prices at R5; **no carbon-price -> sectoral-cost mapping**	§4.1 data; and this absence is *the reason an IO layer exists at all*

Cut from this strand: Nordhaus (2017), redundant with Barrage & Nordhaus, which is itself a DICE paper; and Weitzman (2012), which made the same high-temperature criticism as Pindyck in different vocabulary. Weitzman is the sharper of the two on the quadratic specifically — swap them if §5 is reworked.

Edit note: the NGFS row is doing double duty — it is both a data source and the argument for the method. Possibly split.

D. Four alternative ways to translate a scenario into an economy

Milestone reached: IO is one of four choices, and each omission has a known, signed cost.

Work	Point taken	Where it bites
Hafner et al. (2020)	Classification: E-CGE / macroeconometric / E-SFC / E-AB	§2 framing of the choice
Böhringer et al. (2012)	CGE exemplar. Cross-model spread driven by **substitution elasticities** — invisible in an IO table	§5: fixed coefficients = no substitution = upper bound on transition cost

Mercure et al. (2018)	Macroeconometric (E3ME). \$1–4tn stranded assets — a result requiring the economy **not to clear**	§5: our inelastic-demand assumption
Dafermos, Nikolaidi & Galanis (2018)	E-SFC. Damage -> defaults -> asset deflation -> **feeds back to worsen damage**	§5: we have no financial feedback loop
Lamperti et al. (2018)	E-AB. Damages >> standard IAMs; smoothness of the damage function matters more than its coefficient	§5: challenges our Ω directly

Milestone statement to write: three of four literatures point the bias the same way — *toward understatement*.

E. The input–output tradition

Milestone reached: IO can carry emissions, and its inverse is an economic statement, not a computational trick.

Work	Point taken	Where it bites
Leontief (1936)	The system: $x = Ax + f$, hence the Leontief inverse	§3.2
Hawkins & Simon (1949)	Existence condition via leading principal minors of $I-A$ -> "productive economy"	§3.2 spectral bound; Appendix A
Leontief (1970)	**Pollutants as rows of the accounting system** — origin of environmentally extended IO	§3.3 — every carbon intensity descends from this
Miller & Blair (2022)	Three things: the price dual; the inter-country block structure; **aggregation is not neutral**	§3.2 dual + blocks; §4.2 (aggregation bias is why regions are derived, not asserted)

Edit note: Leontief (1970) is the single most important addition from the last revision — it was previously missing entirely.

F. Multi-regional IO: the data problem, and its dissolution

Milestone reached: the flows that once had to be estimated are now measured, which removes an entire layer of error.

Work	Point taken	Where it bites
Isard (1951)	Exact interregional formulation — and unusable, needs a full bilateral flow matrix	§2 sets up the problem

Chenery (1953) / Moses (1955)	Keep accounting exact, weaken behaviour: one trade coefficient per commodity for all using sectors	§2 — the alternative we don't need
Leontief & Strout (1963)	Estimate the missing flows by gravity	§2 — the alternative we don't need
Stone (1961)	Biproportional fitting: how the gravity frictions are solved	§2
Tukker & Dietzenbacher (2013)	The 2000s programme that *measured* global MRIO tables	§2 turning point
Timmer et al. (2015)	WIOD — the main alternative database (43 economies, ends 2014)	§4.1 database choice
Moran & Wood (2014)	Cross-database carbon accounts agree within ~10% after harmonisation	§4.1 — reassurance, not equivalence
OECD (2025)	The table used: 81 economies, off-diagonal blocks supplied directly	§4.1
Davis & Caldeira (2010)	Consumption-based accounting: much of developed-economy consumption is produced elsewhere -> **trade-weight ranking != embodied-emissions ranking**	§4.2 — the entire justification for the carbon-linkage measure

Edit note: this is the longest strand (9 works). Candidate for trimming if the chapter runs long — Stone could go.

G. Pricing carbon, and who actually pays

Milestone reached: a price can be set but its incidence cannot be derived from an IO table — pass-through must be a swept parameter.

Work	Point taken	Where it bites
Weitzman (1974)	Prices vs quantities: instrument choice depends on relative slopes	§2 background only
Green (2021)	Ex post, measured reductions modest -> a carbon price is a **contested policy variable**	§3.1 — supports treating the scenario as a random variable
Kay & Jolley (2023)	IO price model + NGFS scenarios, **single economy**: 10–30% price rises in carbon-intensive industries at \$200/t	§2 — the closest antecedent; same method, same scenarios, one country

Roncalli & Semet (2024)	Cost-push model; separates own-emissions charge from embodied charge; **indirect burden often larger**; incidence genuinely open	§3.3 direct vs propagated term
RBB Economics (2014)	Pass-through rates vary widely with market structure and shock type	§3.3 — grounds ϕ
Weber & Wasner (2023)	Under general cost pressure, market power -> pass-through **above 1**	§3.3, §4.5 — why the sweep should not stop at ϕ=1
Böhringer et al. (2012) *(reused)*	Leakage 5–20%; BCA reduces but does not eliminate; **a BCA is a tariff differentiated by embodied carbon**	§3.7 — CBAM needs no new machinery

H. From a real shock to a market price

Milestone reached: each link is available off the shelf, but the first one is disputed.

Work	Point taken	Where it bites
Moessner (2022)	\$10/t -> +0.08pp headline inflation, OECD panel 1995–2020	§3.5 the coefficient we adopt
Konradt & Weder di Mauro (2023)	Europe/Canada, 3 decades: effect **indistinguishable from zero**; relative price change, not inflation	§5 — the dispute our point estimate sits inside
Bauer & Känzig (2024)	Carbon pricing moves **expectations** even where realised inflation does not	§5
Taylor (2007)	The monetary reaction function	§3.5
Hull & White (1994)	One-factor: $\Delta R(\tau) = B(\tau)/\tau \cdot \Delta r$, **volatility-independent**	§3.5 — why no vol surface is needed
Sarno & Taylor (2002)	PPP poor at short horizons; CIP close to an arbitrage identity	§3.6 — grades the two FX legs unequally

Edit note: the Moessner/Konradt disagreement is the strongest "honest weakness" in the chapter. Consider giving it its own paragraph rather than folding it in.

I. The framework, and the gap

Work	Point taken	Where it bites
Kenyon, Macrina & Berrahoui (2022)	Provenance: pricing carbon consistently inside financial instruments	§2 — shows BKMN is not a one-off
Berrahoui et al. (2025)	The ensemble: short chain, individually auditable, two proved propositions. Answers Acharya's fourth criticism	§1, §3 throughout

The gap, stated as three existing things and one absence:

1. An auditable stress-testing chain exists — for **one** economy (BKMN).
2. A mature MRIO apparatus exists, with **measured** tables (Miller & Blair; Tukker & Dietzenbacher; OECD).
3. IO carbon-tax incidence exists — for **one** economy (Kay & Jolley; Roncalli & Semet).
4. Nothing joins them. And by Leontief (1970) + Davis & Caldeira (2010), costs demonstrably cross borders; by definition, relative prices do not exist in a one-country model.

Counts

45	Works cited
9	Strands
F (9 works)	Longest strand
1930s–2020s, thinnest at 1980s (0) and 1990s (1)	Decades

Open questions for you

1. **Strand F** is the longest at 9 works. Trim Stone? Or keep — it is the strand that explains why the 1950s–60s references belong at all.
2. **Böhringer (2012)** appears in two strands (D and G). Deliberate — it is both a CGE exemplar and the leakage reference — but flag if it reads as padding.

3. **1980s empty, 1990s = 1.** Fillable with early environmentally-extended IO work between Leontief (1970) and the database era, if the histogram matters.
4. **Should the "where it bites" column survive into the prose?** It is what the supervisor objected to ("you mention a lot of what my work does"). My suggestion: keep it in this outline for your own use, and in the prose let only the closing gap statement refer to the present work.